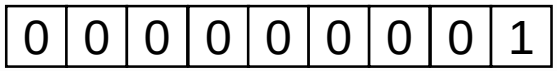
Programowalne Układy Cyfrowe

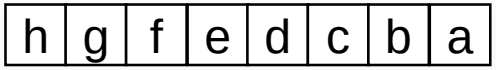
Lab4 - Liczby BCD

Adam Grzelka

2021

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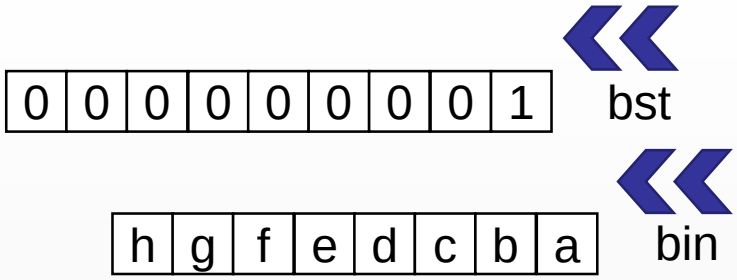





```
reg [31:0] bin;
reg [32:0] bst;

always @(posedge CLK or posedge RST)
if (RST) begin
        bin <= 0;
        bst <= 0;
end else begin
        if (I_STB)
        begin
                bin <= I_DAT;
                bst <= 1;
        end else begin
                bin <= (bin<<1);
                bst <= (bst<<1);
        end
end
```

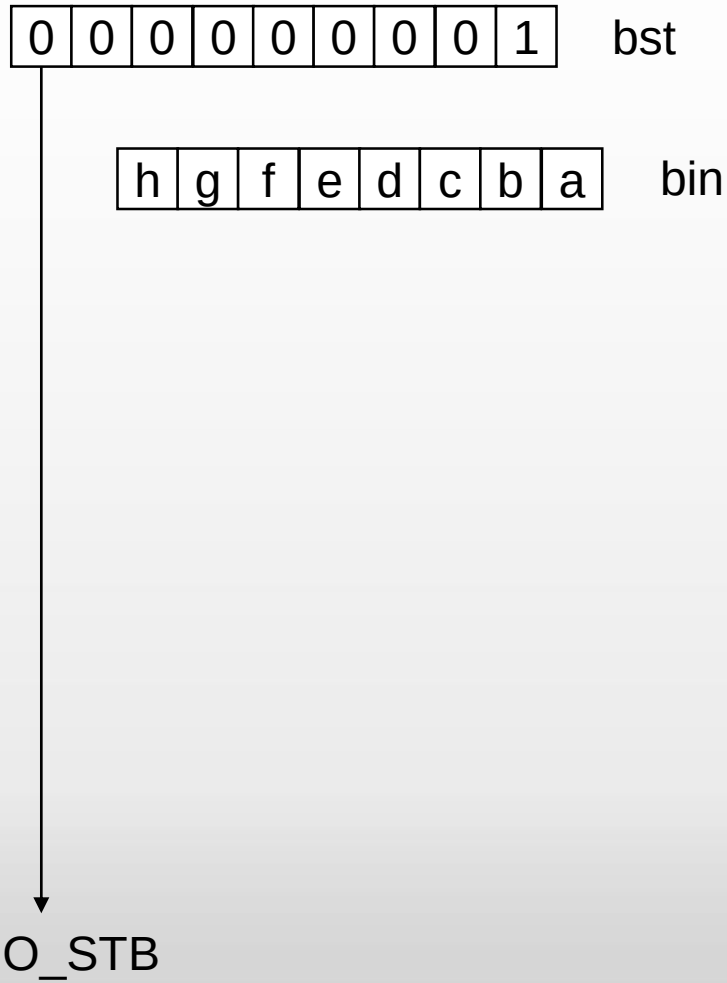
fppt.com




```
reg [31:0] bin;
reg [32:0] bst;

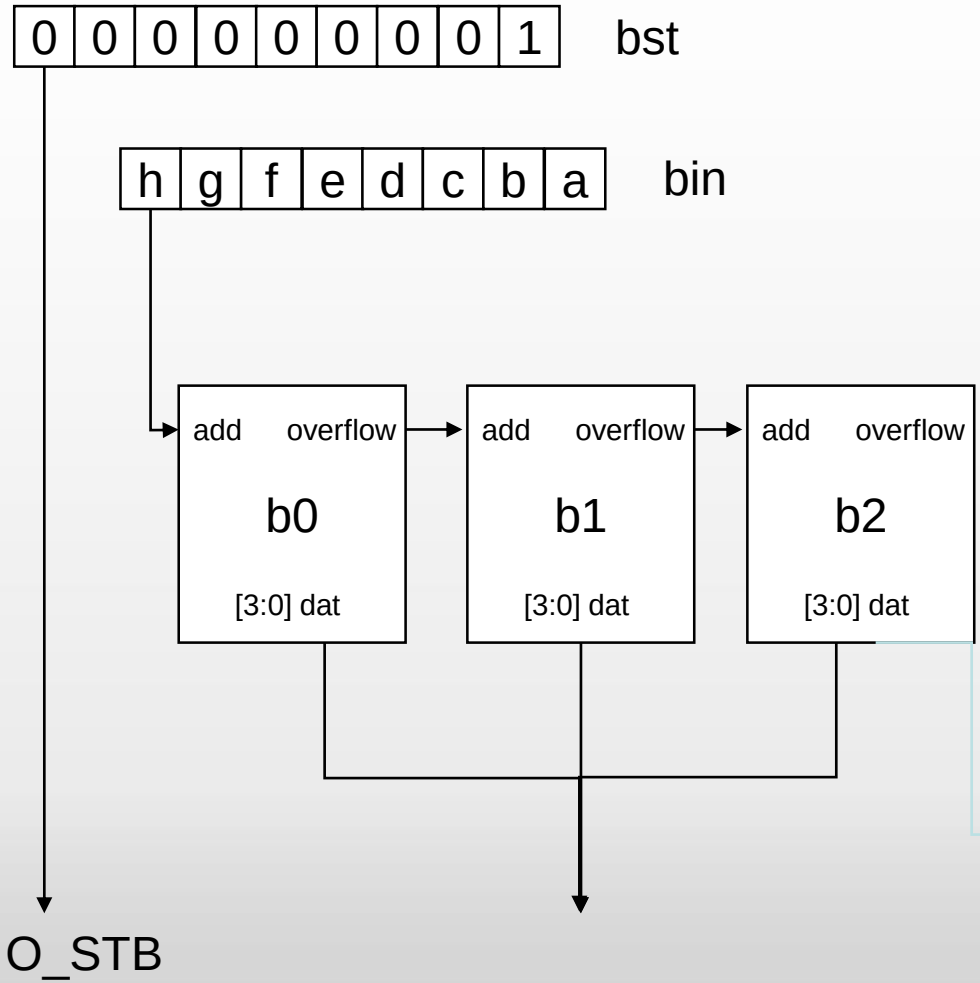
always @(posedge CLK or posedge RST)
if (RST) begin
        bin <= 0;
        bst <= 0;
end else begin
        if (I_STB)
        begin
                bin <= I_DAT;
                bst <= 1;
        end else begin
                bin <= (bin<<1);
                bst <= (bst<<1);
        end
end
```

fppt.com




```
assign O_STB = bst[32];
```

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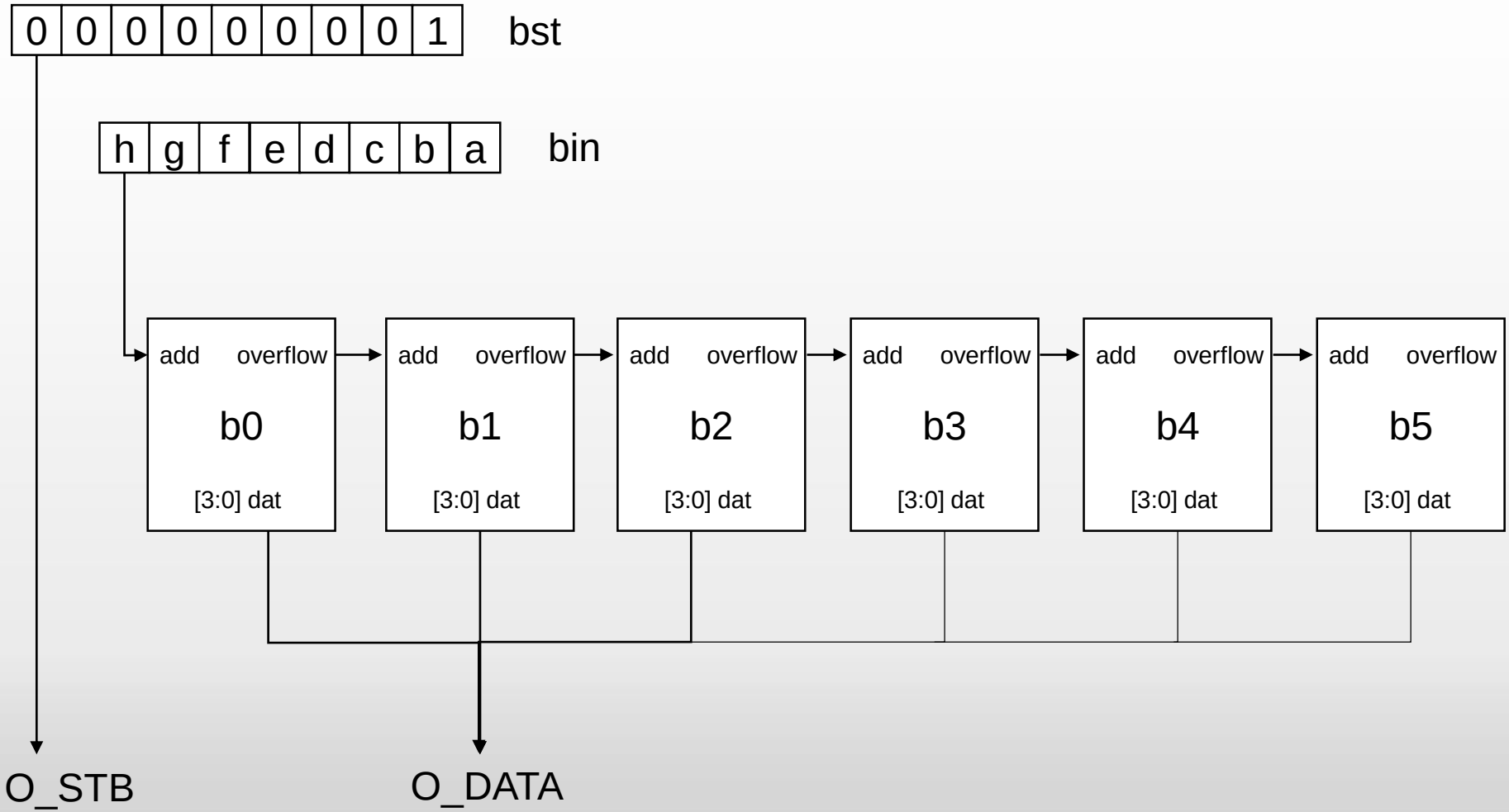



```
bcd_shl_1 b0 (
.CLK(CLK),
.RST(RST),
.ADD1(bin[31]),
.DAT(bcd0),
.OVERFLOW(ovr[0]) );

bcd_shl_1 b1 (
.CLK(CLK),
.RST(RST),
.ADD1(ovr[0]),
.DAT(bcd1),
.OVERFLOW(ovr[1]) );

bcd_shl_1 b2 (
.CLK(CLK),
.RST(RST),
.ADD1(ovr[1]),
.DAT(bcd2),
.OVERFLOW(ovr[2]) );
```

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0 0 0 0 0 0 0 0 1
bst
h g f e d c b a
bin
add overflow
b0
[3:0] dat
add overflow
b1
[3:0] dat
add overflow
b2
[3:0] dat
add overflow
b3
[3:0] dat
add overflow
b4
[3:0] dat
add overflow
b5
[3:0] dat
O_STB
O_DATA

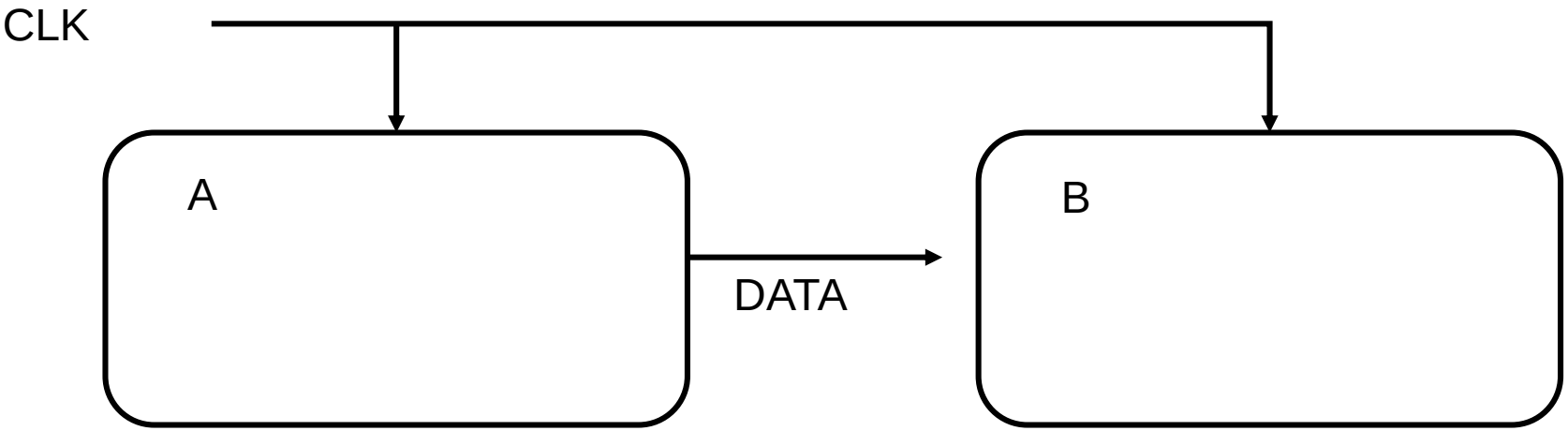
fppt.com

Programowalne Układy Cyfrowe

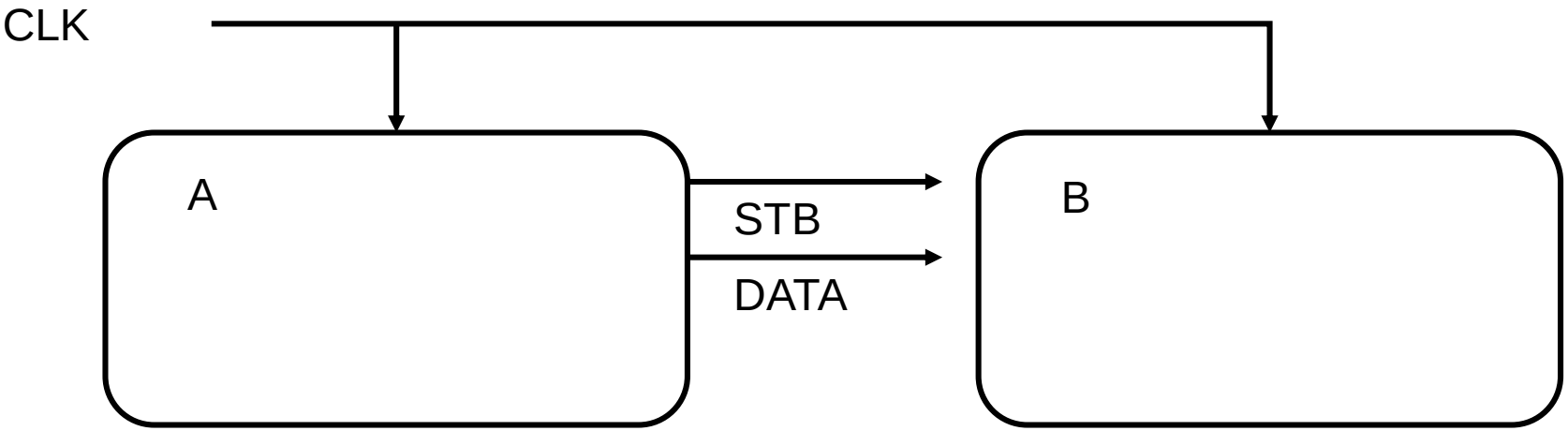
Handshake

fppt.com

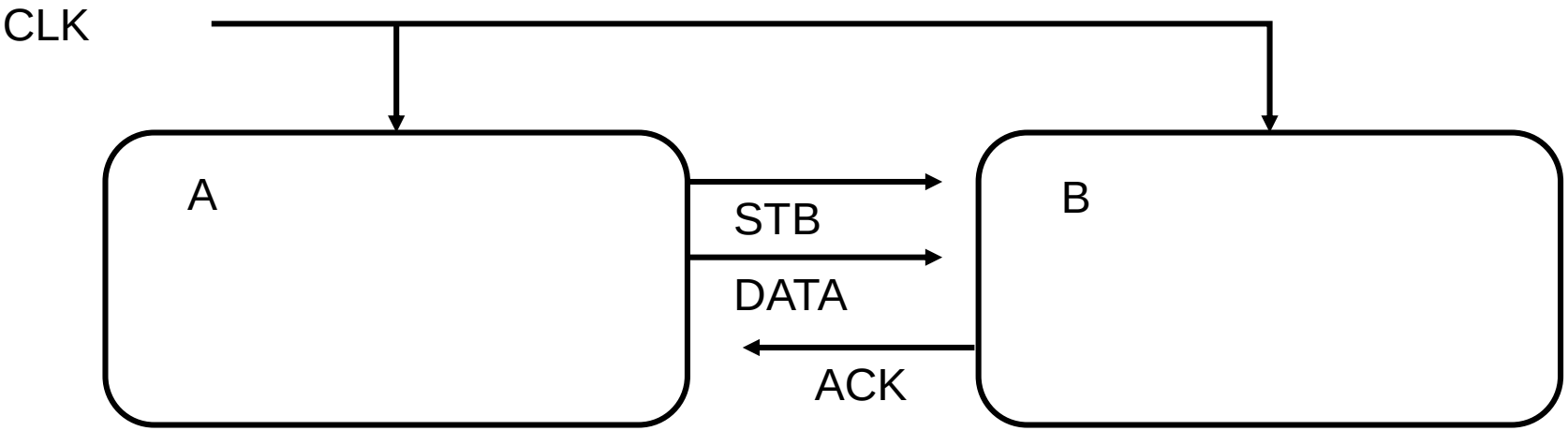
Handshake



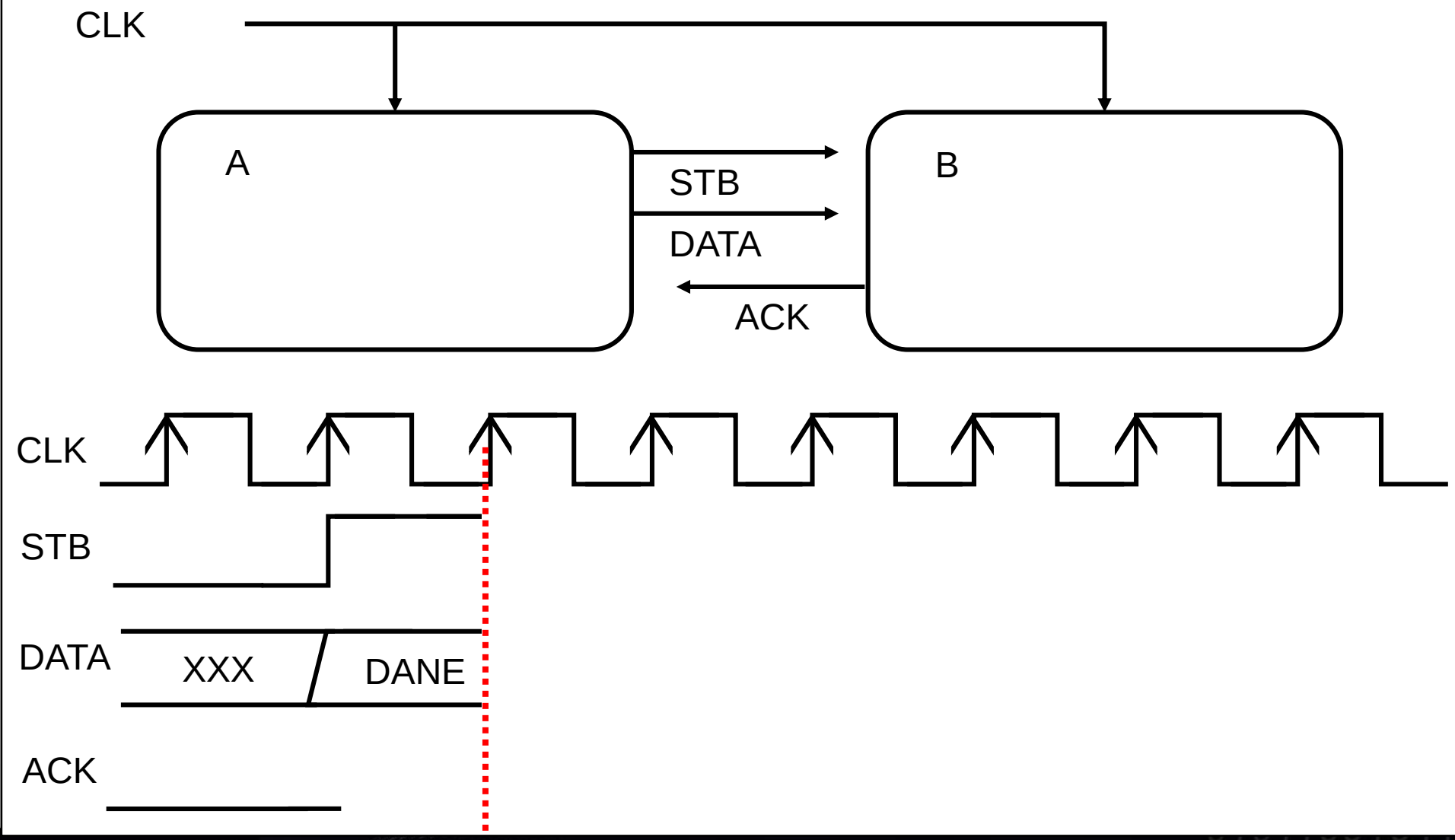
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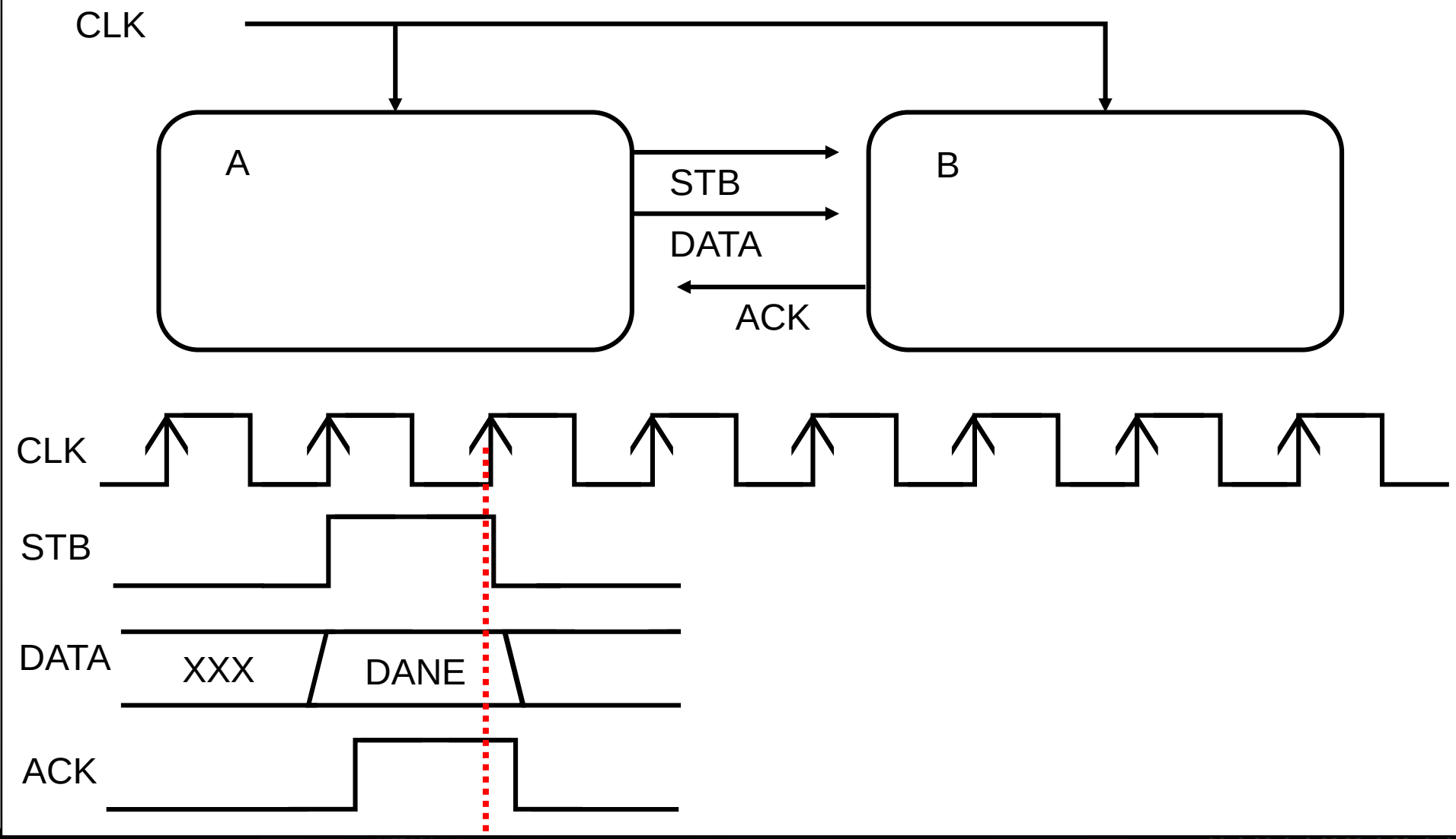
Handshake



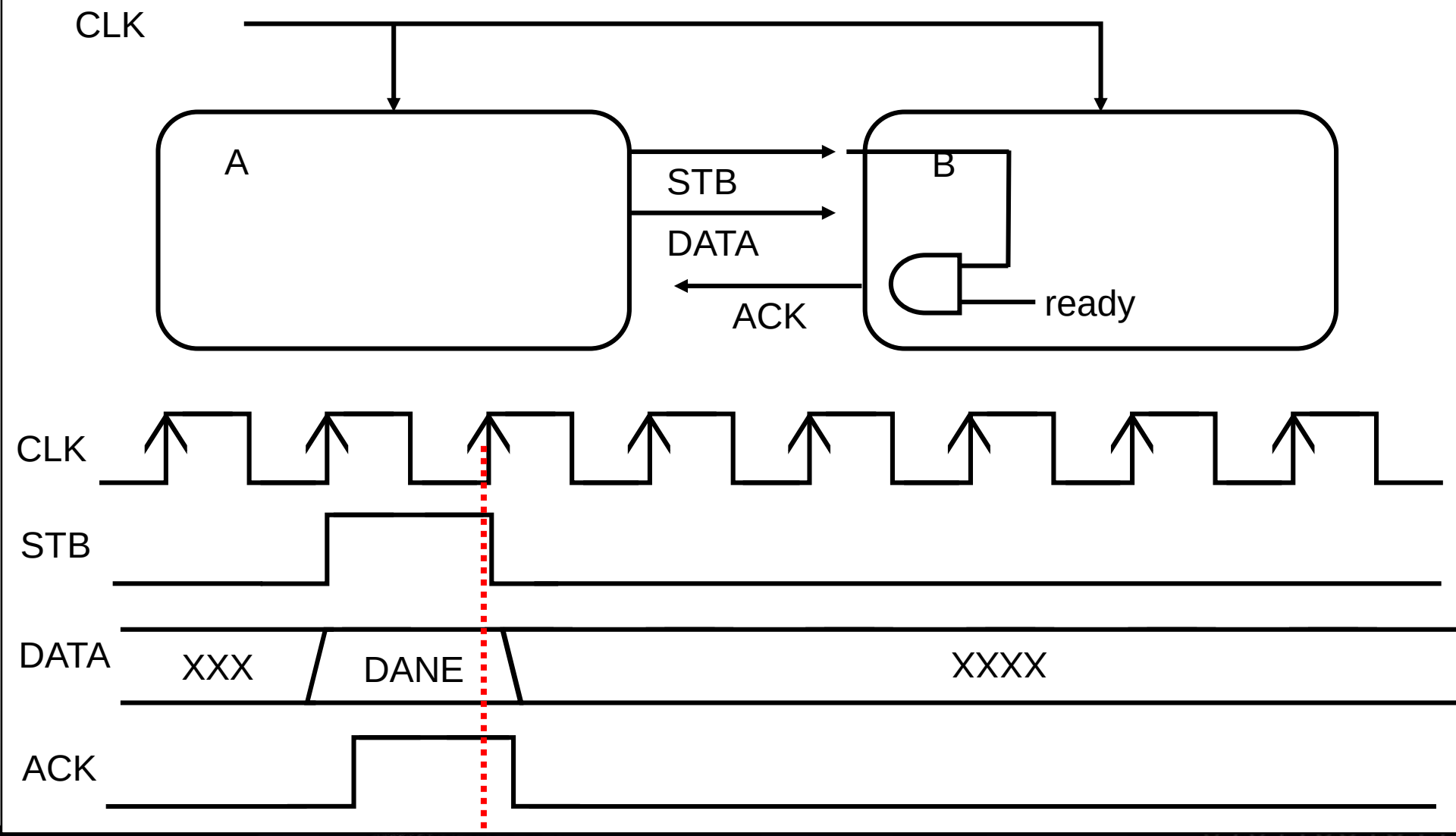
Handshake



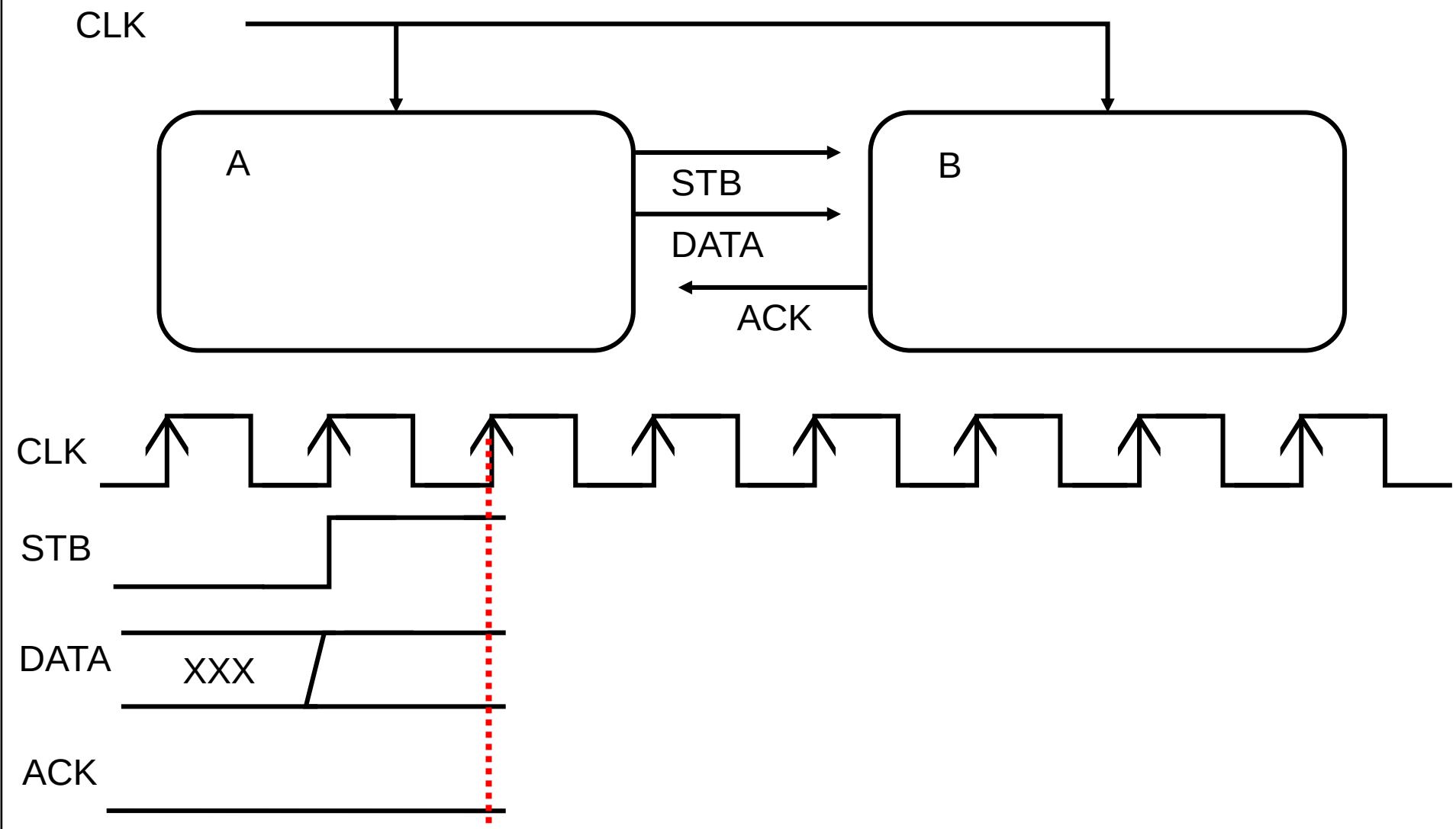
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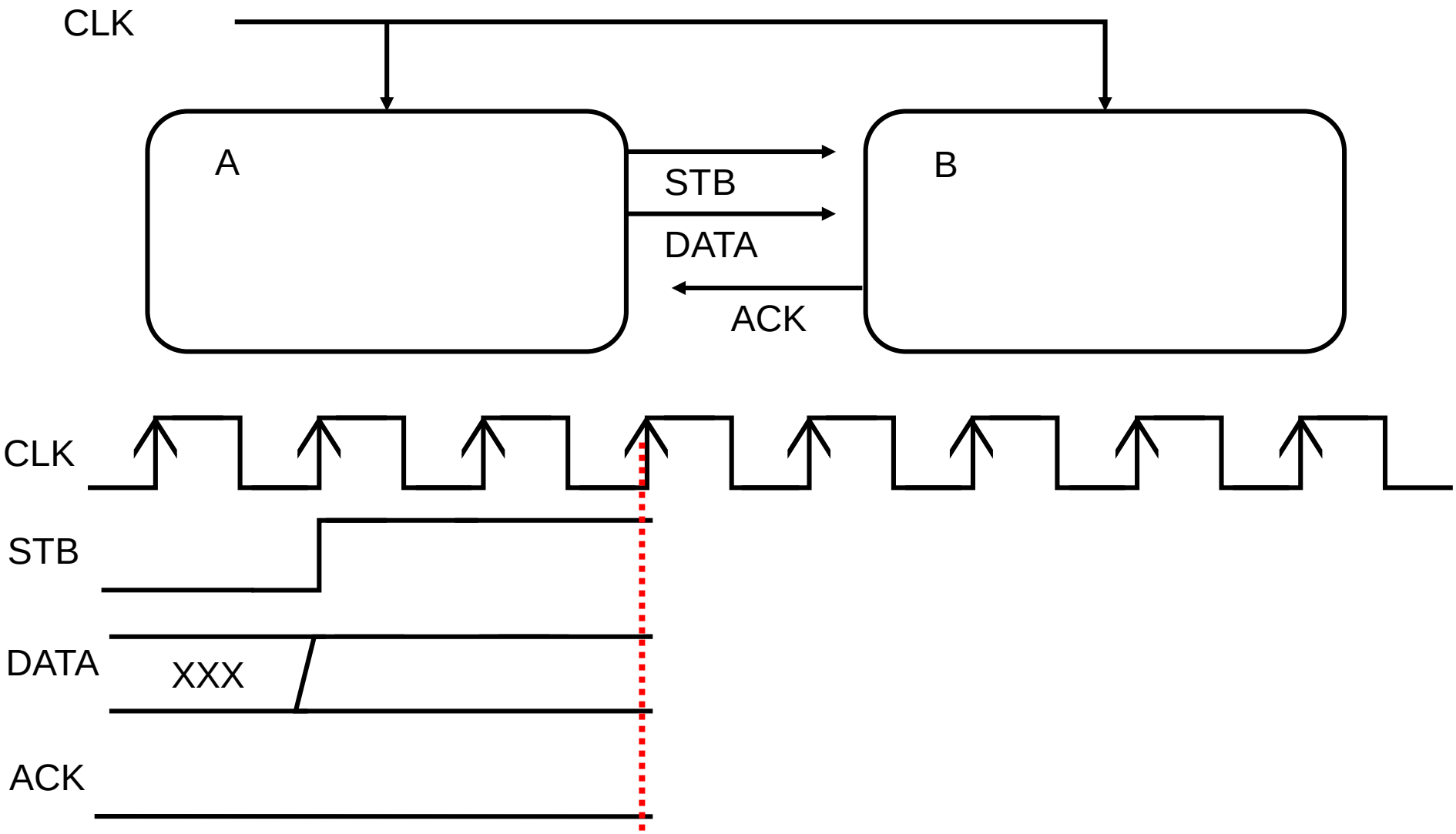
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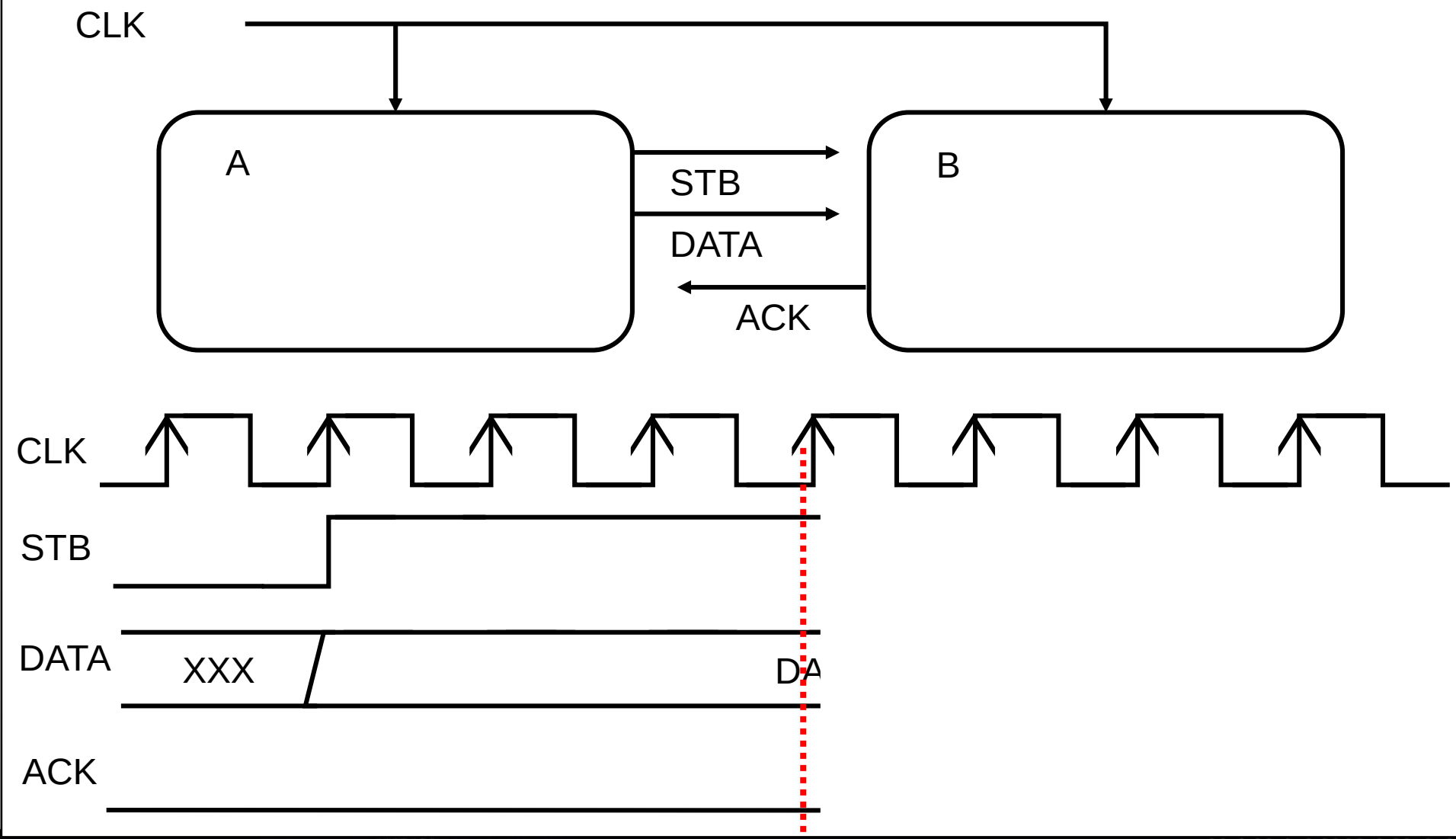
Handshake



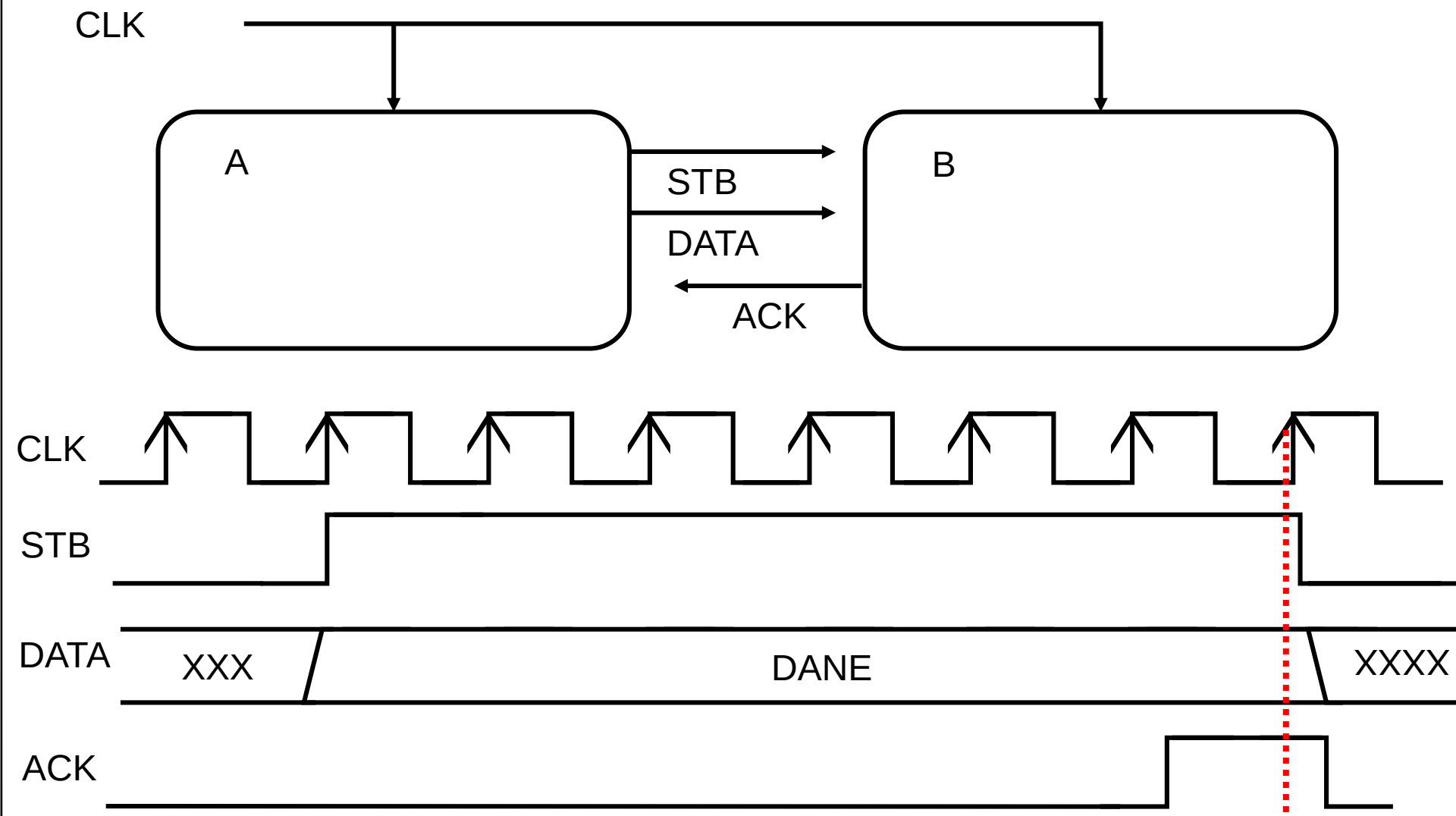
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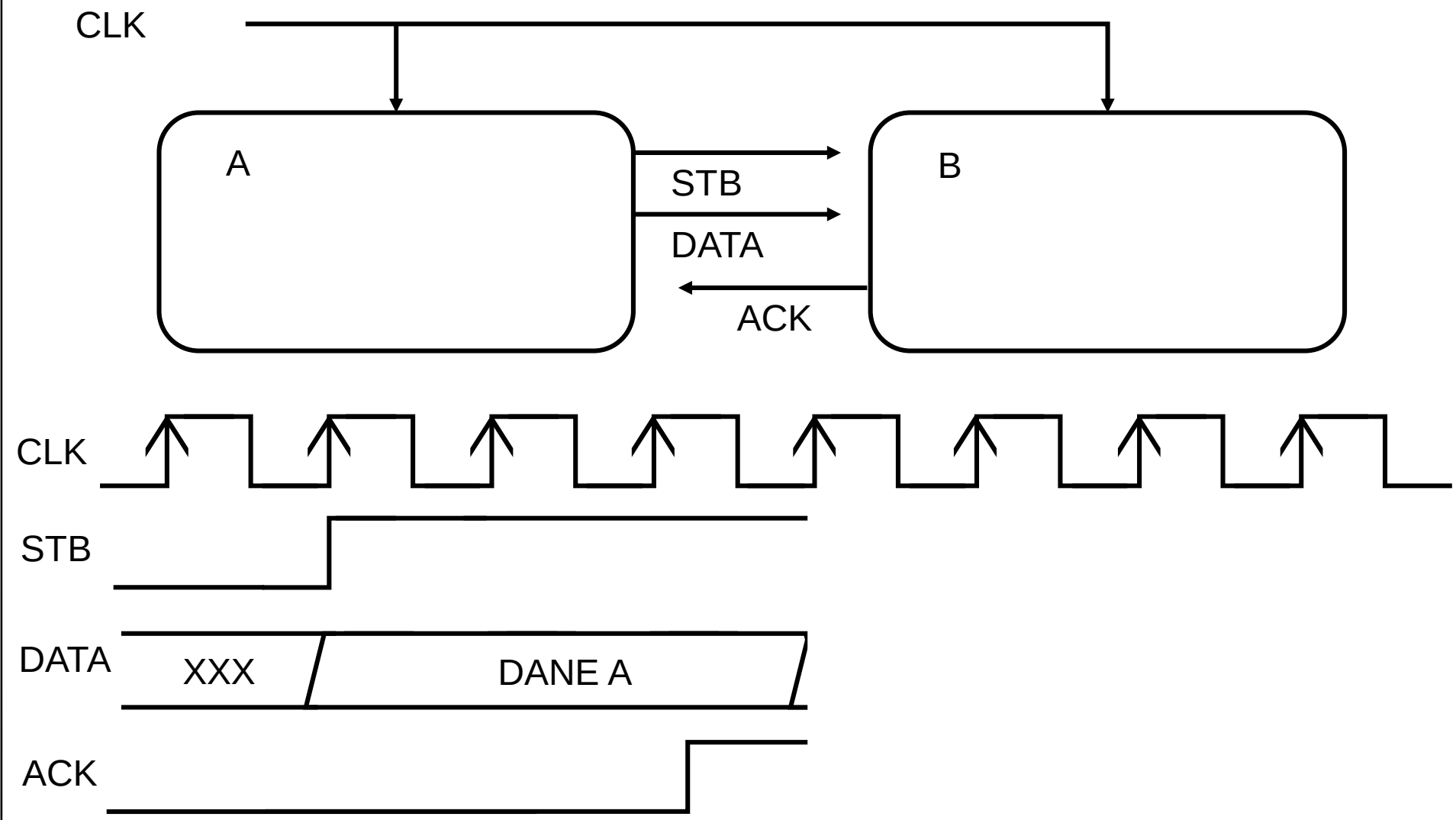
Handshake



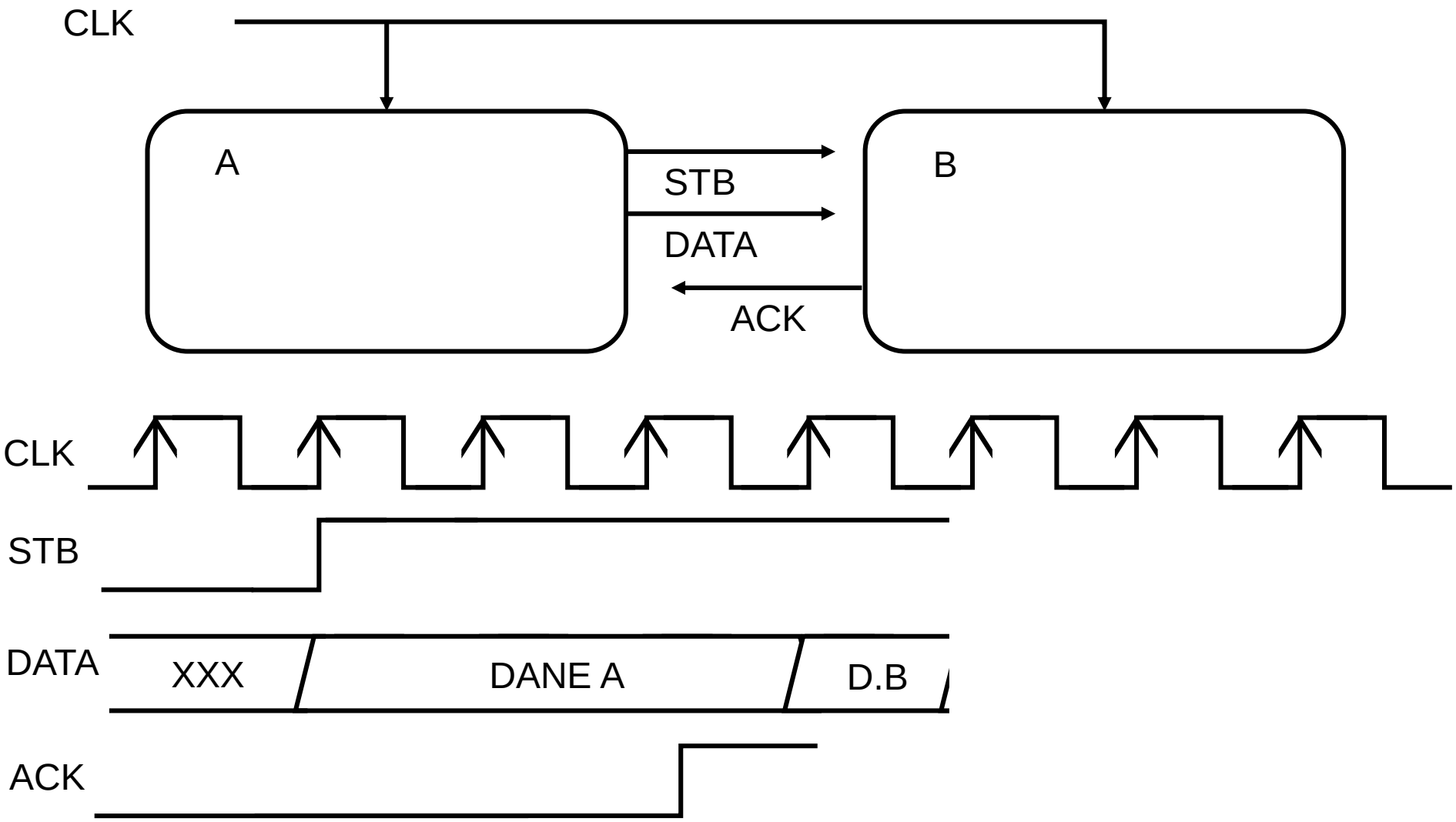
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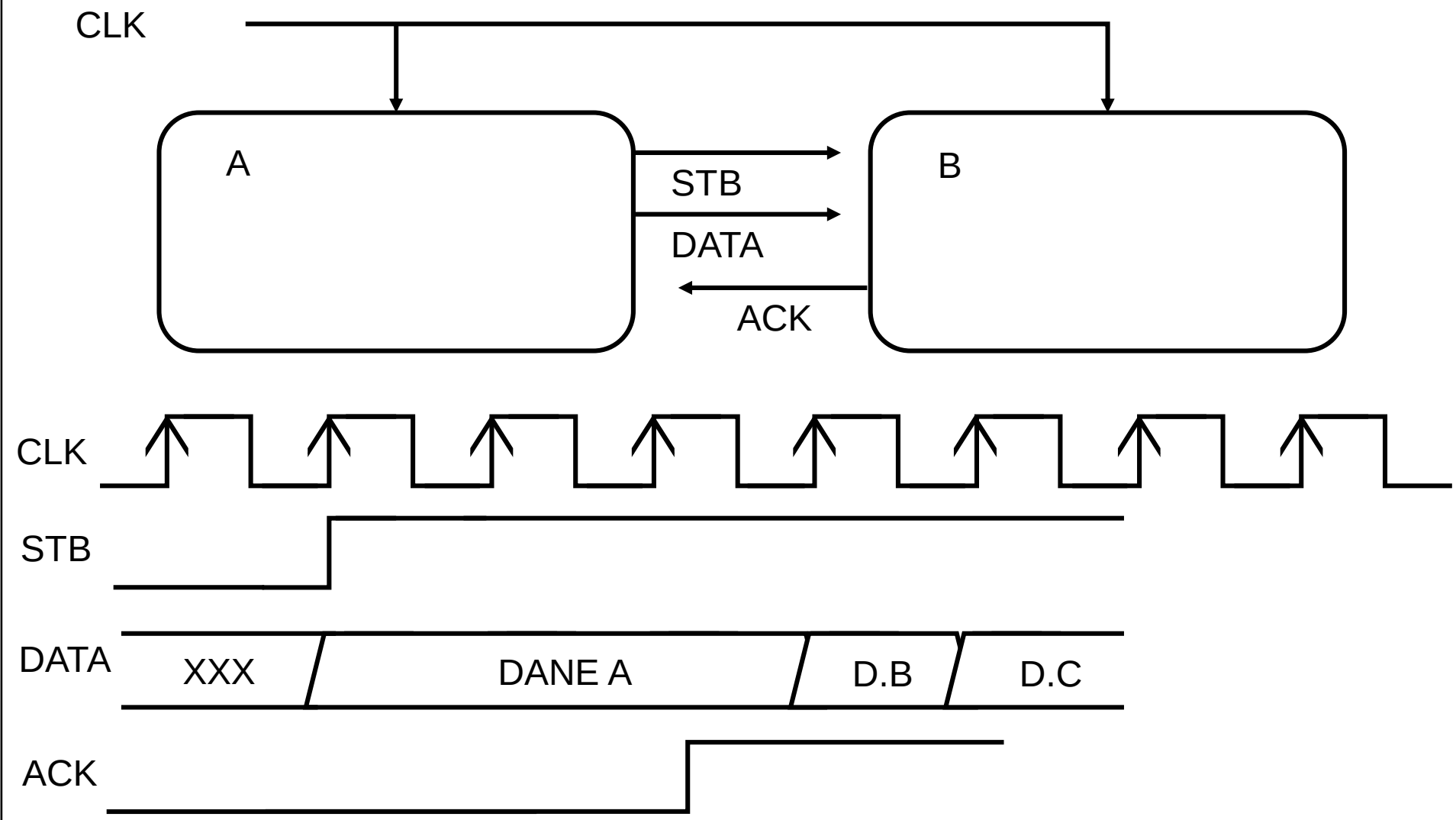
Handshake



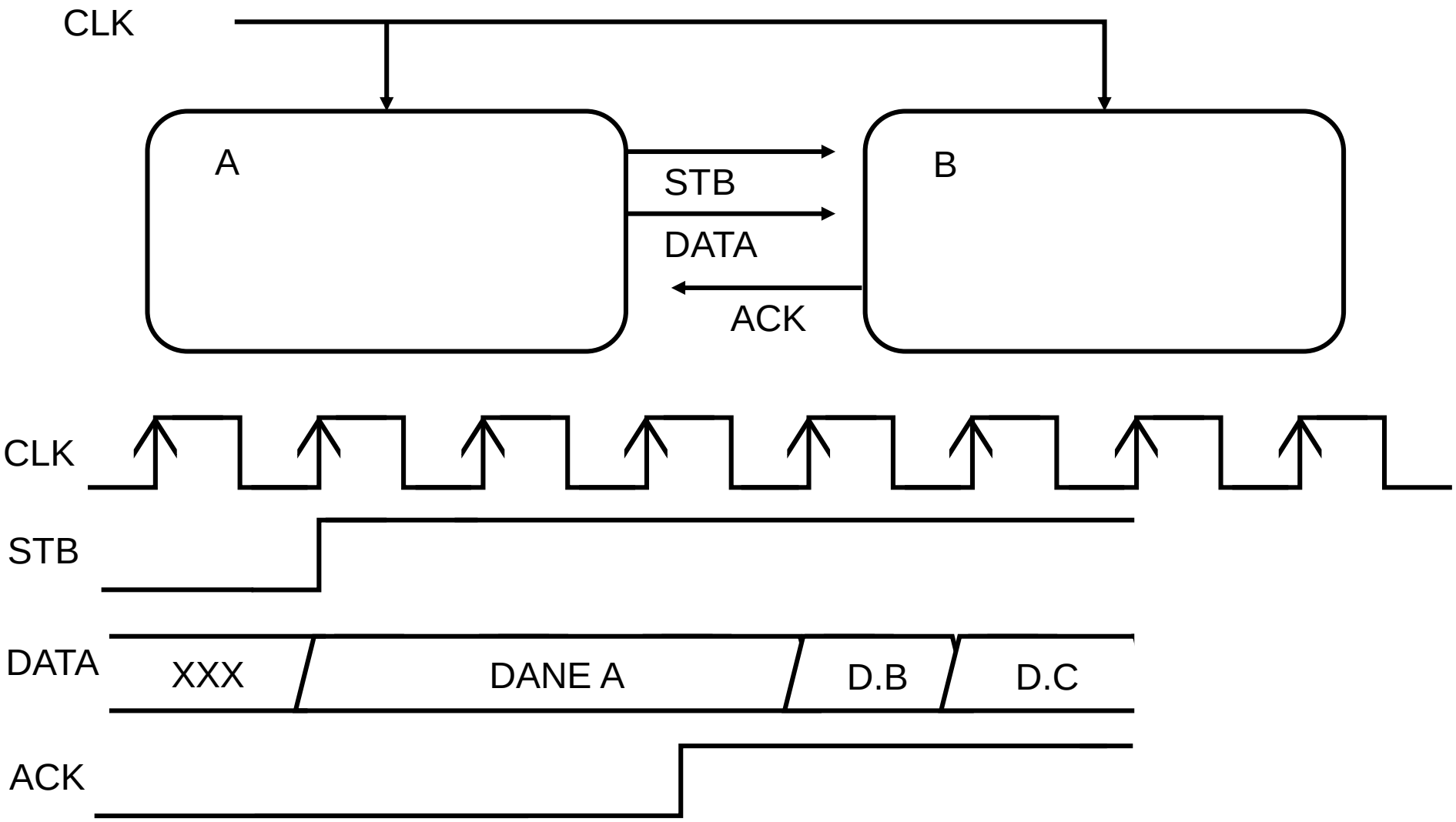
Handshake



Handshake



Handshake



Handshake

